ZACHARY HAREL

CS & Biology Student

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EDUCATION

BS, Computer Science and Biology, Northeastern University, Boston, MA
September 2025 – May 2029 (Expected)

- John Martinson Honors Program Participant
- *Relevant Coursework*: Program Design and Implementation I/II, Discrete Structures, Databases, Computer Systems, Algorithms, Genetics and Molecular Biology, Computational Tools for Genome Analysis

High School Diploma, Half Hollow Hills High School West, Dix Hills, NY
September 2021 – June 2025

- *Academic Research Program*: researched the use of machine learning to aid in diagnoses of diseases including Major Depressive Disorder and Lyme Disease

EXPERIENCE

Machine Learning Intern, EduBeyond, Remote
December 2023 – February 2024

- Created and deployed large language model-backed systems to demonstrate to clients, including use of the ChatGPT API and Flowchat libraries.
- Developed company website features using TypeScript and React.

Teaching Assistant, Program Design and Implementation II, Northeastern University, Boston, MA
September 2026 – Present

- Facilitated weekly recitation sections, including assisting and mentoring students in Java environment and code problems, and leading in-person "codewalk" sessions to verify student understanding of their work.
- Led office hours sessions to provide additional programming support to students and answered student questions through the course discussion platform.

FIRST Technical Advisor (Volunteer), New England FIRST and FIRST Long Island
November 2025 – Present

- Planned, set up, and ran FIRST Tech Challenge qualifiers and regional championship tournaments for over 135 middle and high-school robotics teams across Massachusetts and Long Island.
- Provided support to teams experiencing technical difficulties, particularly with software and electrical issues.
- Provided training and support to other technical volunteers.

OPEN-SOURCE

Co-Lead Maintainer, FRC and FTC Software
May 2026 - Present

- Led infrastructure and curriculum development for frcsoftware.org, an open-source guide and tutorial for programming FIRST Robotics Competition robots.

Contributor, WPILib
August 2025 – Present

- Contributed kinematics, motion planning, and robot operation code for the FIRST Robotics Competition and FIRST Tech Challenge standard library.

CAMPUS INVOLVEMENT

Assistant Command/Data Handling Lead and Website Lead, Northeastern Satellite Laboratory
January 2026 – Present

- Developed flight software for small satellites using the F Prime ecosystem, primarily focusing on board-to-board communication using CAN.
- Developed club logo, branding standards, and website for communication with other labs, clubs, universities, and sponsors.

Assistant Music Director (Fall 2025, Spring 2026), Pit Orchestra (Summer 2026), NUStage Musical Theater Company
September 2025 – Present

- Directed music for two shows in an assistant capacity, including analyzing scores across multiple genres, leading rehearsals for cast members of varying vocal types and skill levels, and arranging harmonies.
- Balanced line-by-line audio levels during performances to ensure an engaging and clear experience for audiences.

TECHNICAL SKILLS

Languages: Python (TensorFlow, PyTorch, NumPy), Java, Kotlin, JavaScript, TypeScript (Svelte, Astro), C, C++, SQL (SQLite, MySQL, PostgreSQL)
Tools: Unix, Git, GitHub, Microsoft 365, Google Workspace, Cloudflare
Domains: Data Analysis, Controls, Machine Learning